Lumina II

SHA-66HD 600-620N

N-Type Bifacial Dual Glass Module
(Hail Resistance Module)

620W
Maximum
Power Output

22.95%
Maximum
Module Efficiency

Super Power Output

SolarSpace advanced N-Type cells combined with MBB and high-density encapsulation provides ultra-high power output

High Reliability

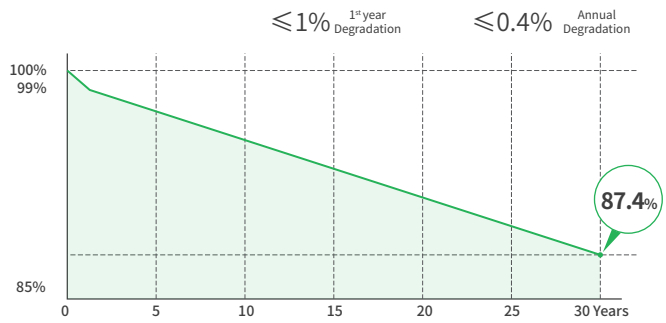
Excellent harsh tests results and advanced half-cell tech improve product reliability for long-term life cycle

Extra power generation

N-type wafers and cells bring ultralow LID&LeTID degradation, less than 1% 1st year degradation guaranteed, in addition lower temperature coefficient and better weak-light response provide extra power generation

High ROI

Bifacial power generation reduces BOS and system LCOE dramatically, promoting the project ROI



12 Years Product Warranty **30** Years Linear Power Warranty

SolarSpace Technology Co., Ltd. was established in 2011, as a world leading solar cell and module manufacturer, concentrating on high efficient solar-technology production with 60GW+ capacity of solar cell and 7.2GW capacity of solar module in China and overseas.

*Please refer to SolarSpace for details

Electric Characteristics（STC）

Module Type	SHA-66HD -600N	SHA-66HD -605N	SHA-66HD -610N	SHA-66HD -615N	SHA-66HD -620N
Maximum Power (Pmax) [W]	600	605	610	615	620
Open-Circuit Voltage (Voc)[V]	48.58	48.78	48.98	49.18	49.38
Maximum Power Voltage (Vmp) [V]	40.47	40.61	40.75	40.89	41.03
Short-Circuit Current (Isc)[A]	15.74	15.80	15.86	15.92	15.98
Maximum Power Current (Imp) [A]	14.84	14.91	14.98	15.05	15.12
Module Efficiency	22.21%	22.40%	22.58%	22.77%	22.95%

Irradiation 1000W/m², Cell Temperature 25°C, AM=1.5

Bifacial Output-Rearside Power Gain (610W)

Power Gain	5%	10%	15%	20%	25%
Maximum Power (Pmax) [W]	641	671	702	732	763
Open-Circuit Voltage (Voc)[V]	48.98	48.98	48.98	49.00	49.00
Maximum Power Voltage (Vmp) [V]	40.75	40.75	40.75	40.77	40.77
Short-Circuit Current (Isc)[A]	16.64	17.46	18.28	19.11	19.93
Maximum Power Current (Imp) [A]	15.74	16.47	17.23	17.96	18.72

Electric Characteristics (NMOT)

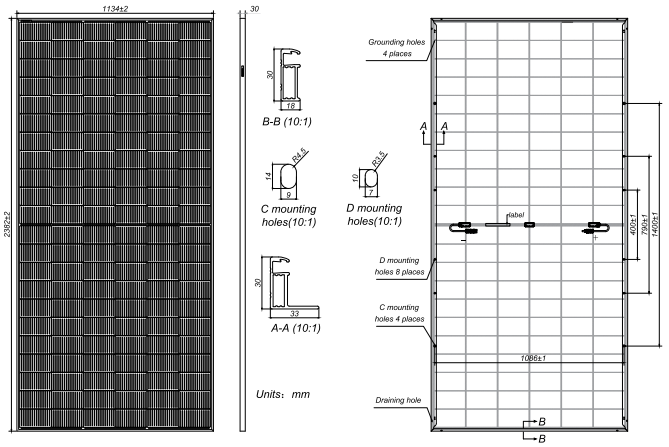
Module Type	SHA-66HD -600N	SHA-66HD -605N	SHA-66HD -610N	SHA-66HD -615N	SHA-66HD -620N
Maximum Power (Pmax) [W]	453	457	461	465	469
Open-Circuit Voltage (Voc)[V]	46.18	46.37	46.56	46.75	46.94
Maximum Power Voltage (Vmp) [V]	37.88	38.05	38.22	38.39	38.55
Short-Circuit Current (Isc)[A]	12.71	12.77	12.83	12.89	12.95
Maximum Power Current (Imp) [A]	11.97	12.02	12.07	12.12	12.17

Irradiance 800 W/m², Ambient Temperature 20 °C, Wind Speed 1 m/s, AM=1.5

Temperature coefficients

Temperature coefficient of Isc	+0.045%/°C
Temperature coefficient of Voc	-0.260%/°C
Temperature coefficient of Pmax	-0.290%/°C
NMOT	45±2°C

Engineering Design



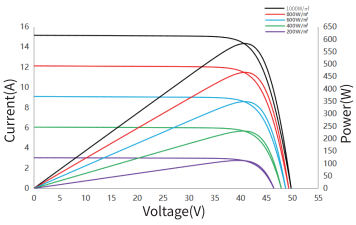
Mechanical Characteristics

Cell Type	Mono N-Type
Number of Cells	132(6x22)
Dimensions	2382X1134X30mm
Weight	40.5kg
Glass	Front glass, 2.5mm coated tempered glass
	Back Glass, 2.5mm tempered glass
Frame	Anodized Aluminum Alloy
Output Cables	4mm²(IEC),12AWG(UL), 300mm (including connector)
Junction Box	IP68 Rated, 3 diodes
Connector	MC4-EVO2 or MC4 Compatible
Packaging	36 Pieces/Pallet, 612 pieces/40' container

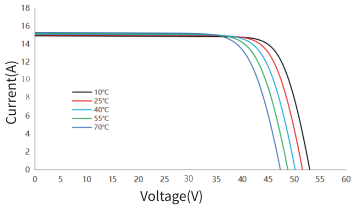
Frame color and cable length are subject to the actual order

Characteristics

I-V/P-V Curve at Different Irradiation
SHA-66HD-610N



I-V Curve at Different Temperature
SHA-66HD-610N



Operating Conditions

Maximum System Voltage	1500V DC (UL)
Power Tolerance	0~+3%
Operating Temperature	-40°C~+85°C
Maximum Series Fuse Rating	30A
Mechanical Load Front Rear	5400Pa
Mechanical Load Back Rear	2400Pa
Hailstone Test	55mm hailstones at 33.9m/s
Bifaciality	80±10%